Perform the following operations using Python on the Facebook metrics data sets a. Create data subsets b. Merge Data c. Sort Data d. Transposing Data e. Shape and reshape Data

In [1]:
```
import pandas as pd
```

In [5]:
```
df = pd.read_csv('dataset_Facebook.csv',delimiter=',')
```

In [6]:
```
df
```

Out[6]:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	P Im
0	0	139441	Photo	2	12	4	3	0.0	2752	
1	1	139441	Status	2	12	3	10	0.0	10460	
2	2	139441	Photo	3	12	3	3	0.0	2413	
3	3	139441	Photo	2	12	2	10	1.0	50128	
4	4	139441	Photo	2	12	2	3	0.0	7244	
...	...	...	...	...	...	...	...	...	...	
495	495	85093	Photo	3	1	7	2	0.0	4684	
496	496	81370	Photo	2	1	5	8	0.0	3480	
497	497	81370	Photo	1	1	5	2	0.0	3778	
498	498	81370	Photo	3	1	4	11	0.0	4156	
499	499	81370	Photo	2	1	4	4	NaN	4188	

500 rows × 20 columns

In [7]:
```
df.describe(include='all')
```

Out[7]:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	Post Ho
count	500.000000	500.000000	500	500.000000	500.000000	500.000000	500.0000
unique	NaN	NaN	4	NaN	NaN	NaN	N
top	NaN	NaN	Photo	NaN	NaN	NaN	N
freq	NaN	NaN	426	NaN	NaN	NaN	N
mean	249.500000	123194.176000	NaN	1.880000	7.038000	4.150000	7.8400
std	144.481833	16272.813214	NaN	0.852675	3.307936	2.030701	4.3685
min	0.000000	81370.000000	NaN	1.000000	1.000000	1.000000	1.0000
25%	124.750000	112676.000000	NaN	1.000000	4.000000	2.000000	3.0000
50%	249.500000	129600.000000	NaN	2.000000	7.000000	4.000000	9.0000
75%	374.250000	136393.000000	NaN	3.000000	10.000000	6.000000	11.0000
max	499.000000	139441.000000	NaN	3.000000	12.000000	7.000000	23.0000

In [8]:

```
df_subset_1 = df[['like','share']]
df_subset_1
```

Out[8]:

	like	share
0	79.0	17.0
1	130.0	29.0
2	66.0	14.0
3	1572.0	147.0
4	325.0	49.0
...	...	...
495	53.0	26.0
496	53.0	22.0
497	93.0	18.0
498	91.0	38.0
499	91.0	28.0

500 rows × 2 columns

In [9]:
```
df_subset_2 = df[['comment','Type']]
df_subset_2
```

Out[9]:

	comment	Type
0	4	Photo
1	5	Status
2	0	Photo
3	58	Photo
4	19	Photo
...	...	...
495	5	Photo
496	0	Photo
497	4	Photo
498	7	Photo
499	0	Photo

500 rows × 2 columns

In [10]:
```
merged_data  = pd.merge(df_subset_2, df_subset_1, left_on='comment', right_on= '
merged_data
```

Out[10]:

	comment	Type	like	share
0	4	Photo	4.0	2.0
1	4	Photo	4.0	1.0
2	4	Photo	4.0	0.0
3	4	Photo	4.0	1.0
4	5	Status	5.0	2.0
...	...	...	...	...
1462	0	Photo	0.0	0.0
1463	0	Photo	0.0	0.0
1464	0	Photo	0.0	0.0
1465	0	Photo	0.0	0.0
1466	0	Photo	0.0	0.0

1467 rows × 4 columns

In [11]:
```
# Define a dictionary containing employee data
data1 = {
'key': ['K0', 'K1', 'K2', 'K3'],
'Name':['Jai', 'Princi', 'Gaurav', 'Anuj'],
```

```
'Age':[27, 24, 22, 32],}

data2 = {
'key': ['K0', 'K1', 'K2', 'K3'],
'Address':['Nagpur', 'Kanpur', 'Allahabad', 'Kannuaj'],
'Qualification':['Btech', 'B.A', 'Bcom', 'B.hons']}

data1 = pd.DataFrame(data1)

data2 = pd.DataFrame(data2)

res = pd.merge(data1, data2, on='key')
res
```

Out[11]:

	key	Name	Age	Address	Qualification
0	K0	Jai	27	Nagpur	Btech
1	K1	Princi	24	Kanpur	B.A
2	K2	Gaurav	22	Allahabad	Bcom
3	K3	Anuj	32	Kannuaj	B.hons

In [12]:

```
merged_data.sort_values(by=['like'],ascending=False)
```

Out[12]:

	comment	Type	like	share
1351	146	Photo	146.0	9.0
1352	146	Photo	146.0	15.0
549	144	Photo	144.0	10.0
550	144	Photo	144.0	29.0
711	64	Photo	64.0	19.0
...	...	...	...	...
891	0	Photo	0.0	0.0
892	0	Photo	0.0	0.0
895	0	Photo	0.0	0.0
896	0	Photo	0.0	0.0
1466	0	Photo	0.0	0.0

1467 rows × 4 columns

In [13]:

```
merged_data.transpose()
#method 1
```

Out[13]:

	0	1	2	3	4	5	6	7	8	9	...
comment	4	4	4	4	5	0	0	0	0	0	...
Type	Photo	Photo	Photo	Photo	Status	Photo	Photo	Photo	Photo	Photo	...
like	4.0	4.0	4.0	4.0	5.0	0.0	0.0	0.0	0.0	0.0	...
share	2.0	1.0	0.0	1.0	2.0	0.0	0.0	0.0	0.0	0.0	...

4 rows × 1467 columns

In [14]:

```
merged_data.T
#method 2
```

Out[14]:

	0	1	2	3	4	5	6	7	8	9	...
comment	4	4	4	4	5	0	0	0	0	0	...
Type	Photo	Photo	Photo	Photo	Status	Photo	Photo	Photo	Photo	Photo	...
like	4.0	4.0	4.0	4.0	5.0	0.0	0.0	0.0	0.0	0.0	...
share	2.0	1.0	0.0	1.0	2.0	0.0	0.0	0.0	0.0	0.0	...

4 rows × 1467 columns

In [16]:

```
df
```

Out[16]:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	P Im
0	0	139441	Photo	2	12	4	3	0.0	2752	
1	1	139441	Status	2	12	3	10	0.0	10460	
2	2	139441	Photo	3	12	3	3	0.0	2413	
3	3	139441	Photo	2	12	2	10	1.0	50128	
4	4	139441	Photo	2	12	2	3	0.0	7244	
...	...	...	...	...	...	...	...	...	...	
495	495	85093	Photo	3	1	7	2	0.0	4684	
496	496	81370	Photo	2	1	5	8	0.0	3480	
497	497	81370	Photo	1	1	5	2	0.0	3778	
498	498	81370	Photo	3	1	4	11	0.0	4156	
499	499	81370	Photo	2	1	4	4	NaN	4188	

500 rows × 20 columns

In [17]:
```
df.Type.unique()
```

Out[17]:
```
array(['Photo', 'Status', 'Link', 'Video'], dtype=object)
```

In [18]:
```
# Reshape
pd.melt(df, id_vars =['Type'], value_vars =['comment'])
```

Out[18]:

	Type	variable	value
0	Photo	comment	4
1	Status	comment	5
2	Photo	comment	0
3	Photo	comment	58
4	Photo	comment	19
...	...	...	...
495	Photo	comment	5
496	Photo	comment	0
497	Photo	comment	4
498	Photo	comment	7
499	Photo	comment	0

500 rows × 3 columns

In [19]:

```
# Reshape
df_temp = pd.DataFrame({'foo': ['one', 'one', 'one', 'two', 'two', 'two'],
                        'bar': ['A', 'B', 'C', 'A', 'B', 'C'],
                        'baz': [1, 2, 3, 4, 5, 6],
                        'zoo': ['x', 'y', 'z', 'q', 'w', 't']})
df_temp
```

Out[19]:

	foo	bar	baz	zoo
0	one	A	1	x
1	one	B	2	y
2	one	C	3	z
3	two	A	4	q
4	two	B	5	w
5	two	C	6	t

In [20]:

```
df_temp.pivot(index='foo', columns='bar', values='baz')
```

Out[20]:

bar	A	B	C
foo			
one	1	2	3
two	4	5	6

In [ ]: